

Notes for ‘Connection between areas and derivatives’

Examples and derivations:

1. Suppose that $y = f(x)$ is a continuous function defined over an interval $[a, b]$ where $a < b$. Recall that the Fundamental Theorem states that the area under the curve over this interval is

$$\int_a^b f(x) dx = F(b) - F(a)$$

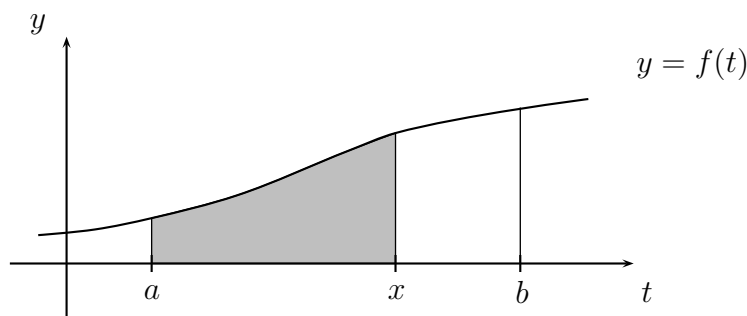
where $F(x)$ is any antiderivative of $f(x)$.

We provide a sketch of a proof of this fact. To simplify the argument, we assume that f is increasing over the interval $[a, b]$.

Sketch of proof: Consider the *area function* A given by the rule

$$A(x) = \int_a^x f(t) dt ,$$

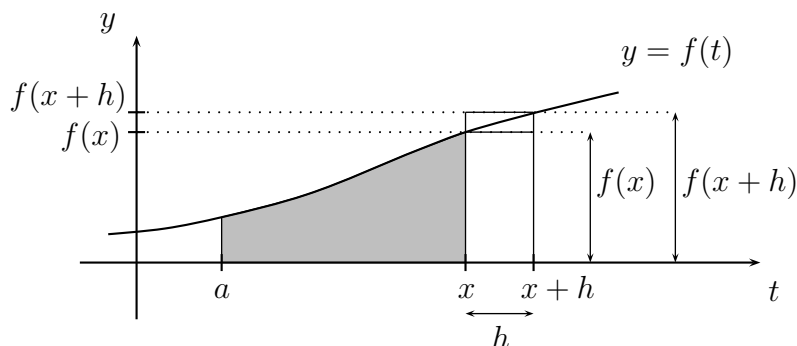
where we have used the variable x as the upper terminal in the definite integral, using instead t for the variable that appears in both the integrand and the differential.



Then $A(x)$ represents the area under the curve $y = f(t)$ over the interval $[a, x]$. In particular,

$$A(a) = \int_a^a f(t) dt = 0 \quad \text{and} \quad A(b) = \int_a^b f(t) dt .$$

Suppose that $a < x < b$ and choose a small positive real number h such that $x < x + h$.



We may draw lower and upper rectangles that enclose the curve $y = f(t)$ for $t \in [x, x + h]$.

On the one hand, the area of the curve over this interval is bounded below by the area of the lower rectangle, which is $hf(x)$. On the other hand, the area of the curve over this interval is bounded above by the area of the upper rectangle, which is $hf(x+h)$. This area is

$$\begin{aligned}\int_x^{x+h} f(t) dt &= \int_x^a f(t) dt + \int_a^{x+h} f(t) dt = \int_a^{x+h} f(t) dt + \int_x^a f(t) dt \\ &= \int_a^{x+h} f(t) dt - \int_a^x f(t) dt = A(x+h) - A(x) .\end{aligned}$$

We thus have the following cascade of inequalities:

$$hf(x) \leq A(x+h) - A(x) \leq hf(x+h) .$$

Dividing through by h , which is positive, retains the inequalities, and we get

$$f(x) \leq \frac{A(x+h) - A(x)}{h} \leq f(x+h) .$$

As $h \rightarrow 0^+$, of course $f(x) \rightarrow f(x)$. But also $f(x+h) \rightarrow f(x)$, since f is continuous. Thus the left and right-hand sides of this cascade of inequalities both tend to the same limit $f(x)$ as $h \rightarrow 0^+$. By the squeeze law,

$$\lim_{h \rightarrow 0^+} \frac{A(x+h) - A(x)}{h} = f(x) .$$

A similar argument, using small negative h and reversed inequalities above, proves that

$$\lim_{h \rightarrow 0^-} \frac{A(x+h) - A(x)}{h} = f(x) .$$

Thus we have the two-sided limit

$$A'(x) = \lim_{h \rightarrow 0} \frac{A(x+h) - A(x)}{h} = f(x) .$$

This proves that $A(x)$ is an antiderivative of $f(x)$.

Suppose now that $F(x)$ is any antiderivative of $f(x)$. Then, there exists a constant C such that

$$F(x) = A(x) + C ,$$

for all $x \in [a, b]$.

(There is some subtlety about this equation at the endpoints of the interval, but this is taken care of by the fact that f is continuous.)

To find C , observe that

$$F(a) = A(a) + C = 0 + C = C .$$

so $C = F(a)$, and the rule for F becomes

$$F(x) = A(x) + F(a) .$$

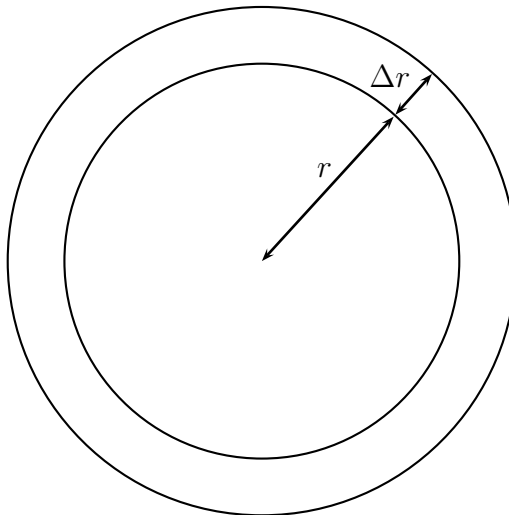
In particular, $F(b) = A(b) + F(a)$. Rearranging this, we get

$$\int_a^b f(t) dt = A(b) = F(b) - F(a) ,$$

completing the sketch of the proof of the Fundamental Theorem of Calculus.

2. We will use the idea of the previous proof to explain why the derivative of the formula for the area of a circle with respect to its radius should produce the formula for the perimeter (without knowing either explicitly).

Let $A = A(r)$ and $P = P(r)$ denote the area and perimeter respectively of a circle of radius r , and consider a small positive change Δr in the radius r , which propagates a small positive change ΔA in the area.



The annulus whose area is represented by ΔA has uniform width Δr throughout, a smaller perimeter $P(r)$ for the inner circle, and a larger perimeter $P(r + \Delta r)$ for the outer circle. Hence the area of the annulus must be bounded below by $P(r)\Delta r$ and bounded above by $P(r + \Delta r)\Delta r$, producing the following cascade of inequalities:

$$P(r)\Delta r \leq \Delta A \leq P(r + \Delta r)\Delta r .$$

Dividing through by Δr , which is positive, retains the inequalities, and we get

$$P(r) \leq \frac{\Delta A}{\Delta r} \leq P(r + \Delta r) .$$

As $\Delta r \rightarrow 0^+$, certainly $P(r) \rightarrow P(r)$, and also $P(r + \Delta r) \rightarrow P(r)$, by continuity of the perimeter function. By the squeeze law

$$\lim_{\Delta r \rightarrow 0^+} \frac{\Delta A}{\Delta r} = P(r) .$$

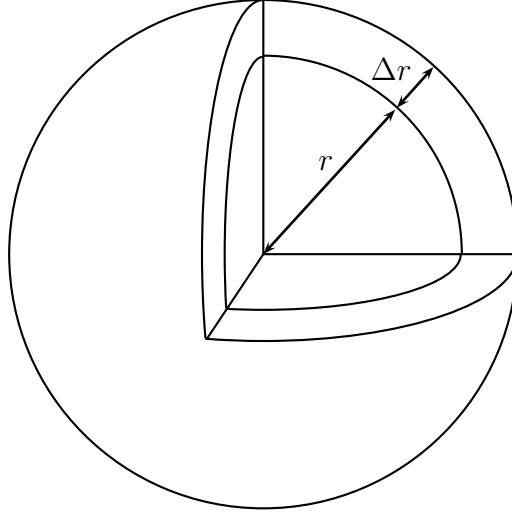
But this limit on the left-hand side is just the derivative (even though expressed here as a one-sided limit), so that

$$\frac{dA}{dr} = P ,$$

as expected. Of course, this matches behaviour we have already observed with regard to explicit formulae for the area and perimeter of a circle.

3. We can naturally extend the idea of the previous example to demonstrate that the derivative of a formula for the volume of a sphere with respect to its radius should produce a formula for the surface area (without knowing either explicitly).

Let $V = V(r)$ and $S = S(r)$ denote the volume and surface area respectively of a sphere of radius r , and consider a small positive change Δr in the radius r , which propagates a small positive change ΔV in the volume.



The “crust” whose volume is represented by ΔV has uniform thickness Δr throughout, all over the sphere. It interfaces with a smaller surface area $S(r)$ for the inner sphere, and a larger surface area $S(r + \Delta r)$ for the outer sphere. Hence the volume of the “crust” must be bounded below by $S(r)\Delta r$ and bounded above by $S(r + \Delta r)\Delta r$, producing the following cascade of inequalities:

$$S(r)\Delta r \leq \Delta V \leq S(r + \Delta r)\Delta r .$$

Dividing through by Δr , which is positive, retains the inequalities, and we get

$$S(r) \leq \frac{\Delta V}{\Delta r} \leq S(r + \Delta r) .$$

As $\Delta r \rightarrow 0^+$, certainly $S(r) \rightarrow S(r)$, and also $S(r + \Delta r) \rightarrow S(r)$, by continuity of the surface area function. By the squeeze law

$$\lim_{\Delta r \rightarrow 0^+} \frac{\Delta V}{\Delta r} = S(r) .$$

But this limit on the left-hand side is just the derivative (even though expressed here as a one-sided limit), so that

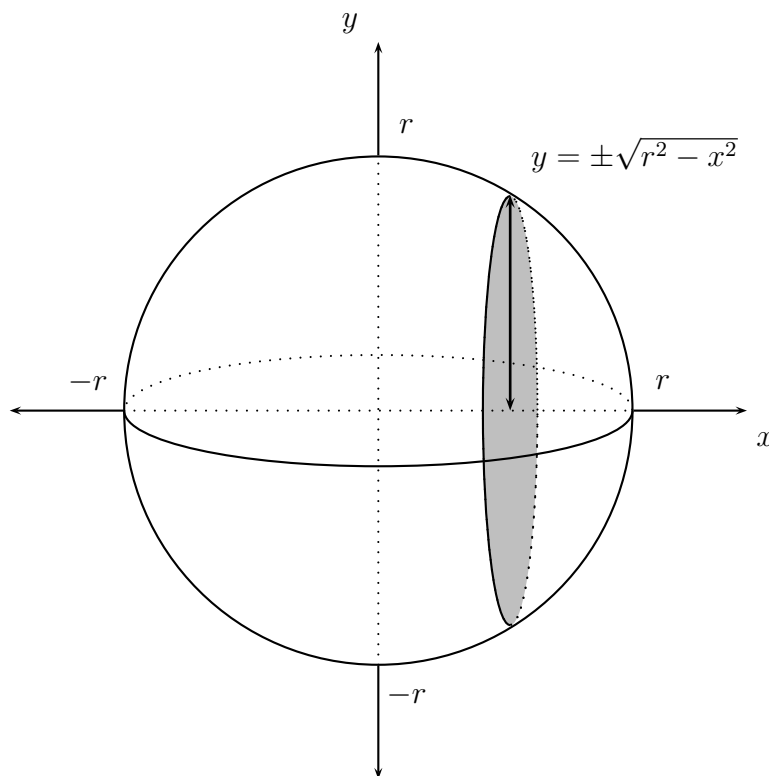
$$\frac{dV}{dr} = S ,$$

as expected.

The upshot of this is that we can deduce a formula for the surface area if we can find an explicit formula for the volume of a sphere, and then just differentiate it with respect to the radius.

4. We now find an explicit formula for the volume of a sphere by integrating its cross-sectional area. The method we are about to use is an instance of what is called the *disk method* in more advanced calculus courses, and we are calculating a *volume of revolution*.

Consider a sphere of radius r positioned in space with the xy -plane forming a vertical cross-section, with the centre of the sphere located at the origin.



The profile of the sphere in the xy -plane is a circle with equation $x^2 + y^2 = r^2$, so that

$$y = \pm\sqrt{r^2 - x^2}.$$

Consider x between $-r$ and r , and form a vertical cross-section of the sphere that is perpendicular to the xy -plane, passing through the x -axis at this particular value of x . The cross-section is a circle with radius $y = \sqrt{r^2 - x^2}$, so its area becomes

$$\pi y^2 = \pi(r^2 - x^2).$$

To get the volume of the sphere, we think of taking a “continuous sum” of these “infinitesimal” vertical cross-sections, as x ranges from $-r$ to r . This idea is intuitive, but it can be formalised using Riemann sums and circular disks of increasingly small widths. Using this idea, the volume of the sphere then becomes

$$\begin{aligned} \int_{-r}^r \pi y^2 dx &= \pi \int_{-r}^r (r^2 - x^2) dx = \pi \left[r^2 x - \frac{x^3}{3} \right]_{-r}^r = \pi \left(r^3 - \frac{r^3}{3} - r^2(-r) + \frac{(-r)^3}{3} \right) \\ &= \pi \left(2r^3 - \frac{2r^3}{3} \right) = \frac{4\pi r^3}{3}, \end{aligned}$$

recovering (discovering!) the usual formula

$$V = \frac{4\pi r^3}{3},$$

for the volume of a sphere. We then recover (discover!) the usual formula for the surface area of a sphere:

$$S = \frac{dV}{dr} = \frac{d}{dr} \left(\frac{4\pi r^3}{3} \right) = 4\pi r^2.$$